

Contemporary Pop Production and the Music Theory of the Electronic Avant-Garde

This briefing document synthesizes the technical methodologies, historical influences, and music theory underpinning modern experimental pop, specifically focusing on the "Brat" aesthetic, the foundations of Bubblegum Bass, and advanced digital processing techniques such as formant shifting and mastering-stage clipping.

Executive Summary

Contemporary experimental pop is characterized by a deliberate tension between "hyper-sweet" mainstream tropes and abrasive, minimalist sound design. The "Brat" aesthetic, popularized by Charli XCX and her collaborators A.G. Cook and George Daniel, represents the commercial apex of the Bubblegum Bass and Hyperpop movements. Technically, this music is defined by a **subtractive production philosophy**, where structure is created by erasing loops rather than adding layers. Vocal processing has evolved from simple pitch correction into a psychological and timbral tool. Auto-Tune is utilized not only for pitch but as a "confidence-builder" in the studio, while **formant shifting** is employed to manipulate vocal "color" and character without altering the fundamental pitch. In mastering, the industry has shifted toward using **clippers before limiters** to achieve aggressive loudness (reaching -6 to -8 LUFS) while preserving the sharp transients essential for dancefloor impact.

The "Brat" Aesthetic: Minimalism and Subtraction

The production of Charli XCX's *Brat* exemplifies a modern shift in Digital Audio Workstation (DAW) workflow. Using Logic Pro and the synthesizer Serum, producers A.G. Cook and George Daniel prioritize sound timbre and "vocal chops" from past recordings to create a cohesive sonic universe.

The Subtractive Philosophy

Unlike traditional analog recording, which is additive (layering drums, bass, guitars, and overdubs), the *Brat* production style is subtractive.

- **Methodology:** Producers start with a saturated arrangement of loops and create structure by deciding when elements exit the mix.
- **Aural Impact:** This results in "ear-grabbingly minimalist" tracks punctuated by abrupt silences and "fake-out" drops that heighten listener attention.
- **Psychological Requirement:** Assertive use of the delete button requires high creative confidence, as it involves erasing work to achieve a cleaner, more impactful result.

Music Theory and Track Analysis

The music of this era often utilizes "exotic modality" and rhythmic disorientation:

- **"360":** Uses rhythmic "fake-outs" where expected anchor notes are replaced by silence. Harmonically, it is ambiguous, jumping between C major, F major, and A minor to keep the listener unsettled.

- **"Sympathy is a Knife":** Incorporates E-flat Lydian mode (noted for its "sharp fourth" A), creating a tense, unstable atmosphere. The track features a "rub" between a pedal B-flat and F chords, mirroring emotional instability.
- **"I Think About It All The Time":** Features a "dystopian sci-fi" sound created through complex chord voicings (e.g., minor seconds, ambiguous fourth stacks) and randomized sound design.

Foundations: Bubblegum Bass and PC Music

Bubblegum Bass is the "hyper-sweet, high-gloss" predecessor to Hyperpop. Emerging in the early 2010s via the London-based PC Music collective (led by A.G. Cook and SOPHIE), it exaggerates mainstream tropes into surrealism.

Core Characteristics

- **Timbres:** Plastic synth sounds, glassy bells, metallic hits, and "supersaws."
- **Vocals:** Heavily pitch-shifted and formant-shifted "childlike" hooks.
- **Imagery:** Consumerist and self-referential, often merging sincerity with satire (e.g., the QT project's energy drink concept).
- **Production:** Heavy sidechaining ("pumping") and OTT (over-the-top) compression to keep the sound "inflated."

Technical Vocal Manipulation

Vocal processing is the cornerstone of the genre, moving beyond simple correction into the realm of identity manipulation.

Formant Shifting vs. Pitch Shifting

While pitch shifting changes the fundamental frequency (creating the "chipmunk effect" if pushed too far), **formant shifting** changes the timbre or "color" of the voice by shifting the harmonics while keeping the fundamental pitch the same. | Technique | Result on Voice || ----- | ----- || **Pitch Shifting** | Changes the note being sung; alters fundamental frequency. || **Formant Shifting** | Changes the "size" of the vocal tract (e.g., making a voice sound more "masculine" or "feminine"). || **Auto-Tune (Robot)** | Locks the vocal to a single note, stripping away natural inflection. |

The Psychological Role of Auto-Tune

For artists like Charli XCX, real-time Auto-Tune in studio headphones acts as a psychological tool. By making it "impossible to sing wrong," it encourages singers to be less inhibited and try experimental melodies they might otherwise avoid.

Historical Context: 1980s Synthesis and "Poke Bass"

Modern production remains heavily indebted to the digital synthesis revolution of the mid-1980s, particularly the techniques found in Scritti Politti's *Cupid & Psyche 85*.

The "Poke Bass" Technique

Developed by David Gamson, this technique is still used to ensure bass sounds cut through complex electronic mixes:

1. **Attack Layer:** A percussive FM synth patch (usually from a Yamaha DX7) with a fast attack, dialed down so only the "click" is heard.
2. **Body Layer:** A fat analog bass sound (typically from a Moog Minimoog).
3. **Result:** A bass that is simultaneously "punchy and cutting" but also "big, round, and solid."

Influential 80s Hardware

- **Fairlight CMI:** Used for gritty 8-bit samples and orchestral embellishments.
- **Yamaha DX7:** Defined the "crystalline sheen" of 80s pop through FM synthesis.
- **PPG Wave:** Provided digital wavetable grit and shining bell tones.
- **AMS DMX 15-80 S:** An early digital delay used to trigger 16-bit drum samples, a precursor to modern DAW sampling.

Mastering for Loudness: Clippers vs. Limiters

In the competitive market of Electronic Dance Music (EDM) and Hyperpop, achieving high integrated loudness is a standard requirement. Engineers frequently use a combination of clippers and limiters on the master bus.

Comparison of Loudness Tools

Tool, Mechanism, Impact on Sound

Limiter, Gain Reduction (turning a fader down), "Cleaner, but can make transients ""soft and dull"" or cause ""pumping.""

Clipper, Distortion (chopping off peaks), "Preserves transient ""snap"" and ""edge,"" but adds audible distortion if pushed."

Recommended Workflow

The most effective chain for "competitive loudness" (-6 to -8 LUFS) is **Clipper -> Limiter**.

1. **Clipper:** Reduces the transient peaks first, which lightens the load on the limiter.
2. **Limiter:** Works on the sustained signal to achieve the final volume without sucking the life out of the transients.
3. **Oversampling:** Often disabled for hard clipping (to preserve headroom) but enabled for soft clipping (to avoid aliasing).

Industry Standards

While streaming services often recommend -14 LUFS, professional mastering engineers for electronic genres note that masters must match the reference tracks of the genre, which are significantly louder (typically -7 LUFS) to remain "competitive" in an over-saturated market.